1. Print numbers from 1 to 10

```
for i in range(1, 11):
    print(i)
```

2. Print even numbers from 1 to 20

```
for i in range(1, 21):
    if i % 2 == 0:
        print(i)
```

3. Multiplication table of a number

```
n = 5
for i in range(1, 11):
    print(f"{n} x {i} = {n*i}")
```

4. Factorial of a number

```
n = 5
fact = 1
for i in range(1, n+1):
    fact *= i
print(fact)
```

5. Fibonacci series

```
n = 10
a, b = 0, 1
for _ in range(n):
    print(a, end=" ")
    a, b = b, a+b
```

6. Check prime number

```
n = 7
for i in range(2, n):
```

```python
    if n % i == 0:
        print("Not Prime")
        break
else:
    print("Prime")
```

7. Reverse a number

```python
n = 123
rev = 0
while n > 0:
    rev = rev*10 + n%10
    n //= 10
print(rev)
```

8. Palindrome number

```python
n = 121
temp = n
rev = 0
while n > 0:
    rev = rev*10 + n%10
    n //= 10

print("Palindrome" if temp == rev else "Not Palindrome")
```

9. Reverse a string

```python
s = "python"
print(s[::-1])
```

10. Palindrome string

```python
s = "madam"
print("Palindrome" if s == s[::-1] else "Not Palindrome")
```

11. Count vowels

```
s = "python"
vowels = "aeiou"
count = 0
for ch in s:
    if ch in vowels:
        count += 1
print(count)
```

12. Count characters frequency

```
s = "aabbcc"
freq = {}
for ch in s:
    freq[ch] = freq.get(ch, 0) + 1
print(freq)
```

13. Find largest number in list

```
lst = [10, 25, 5, 40]
print(max(lst))
```

14. Remove duplicates from list

```
lst = [1, 2, 2, 3, 4, 4]
print(list(set(lst)))
```

15. Sum of list elements

```
lst = [1, 2, 3, 4]
print(sum(lst))
```

16. Second largest number

```
lst = [10, 20, 30, 40]
```

```
lst.sort()
print(lst[-2])
```

17. Armstrong number

```
n = 153
temp = n
s = 0
while temp > 0:
    d = temp % 10
    s += d ** 3
    temp //= 10

print("Armstrong" if s == n else "Not Armstrong")
```

18. GCD of two numbers

```
a, b = 12, 18
while b:
    a, b = b, a % b
print(a)
```

19. Anagram check

```
s1 = "listen"
s2 = "silent"
print("Anagram" if sorted(s1) == sorted(s2) else "Not Anagram")
```

20. Find common elements in two lists

```
l1 = [1,2,3,4]
l2 = [3,4,5]
print(list(set(l1) & set(l2)))
```

21. Count words in a sentence

```python
s = "Python is very easy"
print(len(s.split()))
```

22. Find missing number in array

```python
lst = [1,2,3,5]
n = 5
total = n*(n+1)//2
print(total - sum(lst))
```

23. Move zeros to end

```python
lst = [1,0,2,0,3]
res = [x for x in lst if x != 0] + [0]*lst.count(0)
print(res)
```

24. Pair with given sum

```python
lst = [1,2,3,4,5]
target = 6
for i in range(len(lst)):
    for j in range(i+1, len(lst)):
        if lst[i] + lst[j] == target:
            print(lst[i], lst[j])
```

25. Remove duplicates without using set

```python
lst = [1,2,2,3]
res = []
for i in lst:
    if i not in res:
        res.append(i)
print(res)
```

26. Class and object

```
class Employee:
    def __init__(self, name):
        self.name = name

    def show(self):
        print(self.name)

e = Employee("Rahul")
e.show()
```

27. Inheritance

```
class A:
    def show(self):
        print("Class A")

class B(A):
    pass

obj = B()
obj.show()
```

28. Read file

```
f = open("data.txt", "r")
print(f.read())
f.close()
```

29. Count words in file

```
f = open("data.txt")
print(len(f.read().split()))
```

```
f.close()
```

Print numbers from 1 to N

```
n = 10
for i in range(1, n+1):
    print(i)
```

2 Print even and odd numbers

```
n = 10
for i in range(1, n+1):
    if i % 2 == 0:
        print(i, "Even")
    else:
        print(i, "Odd")
```

3 Sum of N natural numbers

```
n = 5
total = 0
for i in range(1, n+1):
    total += i
print(total)
```

4 Factorial of a number

```
n = 5
fact = 1
for i in range(1, n+1):
    fact *= i
print(fact)
```

5 Fibonacci series

```
n = 7
a, b = 0, 1
for _ in range(n):
    print(a, end=" ")
    a, b = b, a+b
```

6 Check prime number

```
n = 7
if n > 1:
    for i in range(2, n):
        if n % i == 0:
            print("Not Prime")
            break
    else:
        print("Prime")
```

7 Print prime numbers between 1 and 100

```
for n in range(2, 101):
    for i in range(2, n):
        if n % i == 0:
            break
    else:
        print(n, end=" ")
```

8 Reverse a number

```
n = 123
rev = 0
while n > 0:
    rev = rev*10 + n%10
    n //= 10
```

```
print(rev)
```

9 Palindrome number

```
n = 121
temp = n
rev = 0
while n > 0:
    rev = rev*10 + n%10
    n //= 10

print("Palindrome" if temp == rev else "Not Palindrome")
```

10 Armstrong number

```
n = 153
temp = n
s = 0
while temp > 0:
    d = temp % 10
    s += d ** 3
    temp //= 10

print("Armstrong" if s == n else "Not Armstrong")
```

11 Star pattern

```
*
**
***
for i in range(1, 4):
    print("*" * i)
```

12 Number pattern

```
1
12
123
for i in range(1, 4):
    for j in range(1, i+1):
        print(j, end="")
    print()
```

13 Pyramid pattern

```
  *
 ***
*****
n = 3
for i in range(1, n+1):
    print(" "*(n-i) + "*"*(2*i-1))
```

14 Reverse a string

```
s = "python"
print(s[::-1])
```

15 Palindrome string

```
s = "madam"
print("Palindrome" if s == s[::-1] else "Not Palindrome")
```

16 Count vowels in string

```
s = "python"
count = 0
for ch in s:
    if ch in "aeiou":
        count += 1
```

```
print(count)
```

17 Count words in string

```
s = "Python is easy"
print(len(s.split()))
```

18 Count characters in string

```
s = "hello"
print(len(s))
```

19 Find duplicate characters

```
s = "programming"
duplicates = []
for ch in s:
    if s.count(ch) > 1 and ch not in duplicates:
        duplicates.append(ch)
print(duplicates)
```

20 Remove spaces from string

```
s = "hello world"
print(s.replace(" ", ""))
```

21 Largest element in list

```
lst = [10, 20, 5, 40]
print(max(lst))
```

22 Smallest element in list

```
lst = [10, 20, 5, 40]
print(min(lst))
```

23 Sum of list elements

```
lst = [1, 2, 3, 4]
print(sum(lst))
```

24 Remove duplicates from list

```
lst = [1, 2, 2, 3]
print(list(set(lst)))
```

25 Sort list

```
lst = [5, 2, 9, 1]
lst.sort()
print(lst)
```

26 Find second largest element in list

```
lst = [10, 20, 30, 40]
lst = list(set(lst))
lst.sort()
print(lst[-2])
```

27 Count frequency of elements in list

```
lst = [1, 2, 2, 3, 3, 3]
freq = {}

for i in lst:
    freq[i] = freq.get(i, 0) + 1

print(freq)
```

28 Find common elements in two lists

```
l1 = [1, 2, 3, 4]
l2 = [3, 4, 5]
```

```
common = list(set(l1) & set(l2))
print(common)
```

29 Reverse a list

```
lst = [1, 2, 3, 4]
print(lst[::-1])
```

30 Merge two lists

```
l1 = [1, 2]
l2 = [3, 4]
print(l1 + l2)
```

31 Check anagram strings

```
s1 = "listen"
s2 = "silent"

print("Anagram" if sorted(s1) == sorted(s2) else "Not Anagram")
```

32 First non-repeated character

```
s = "swiss"
for ch in s:
    if s.count(ch) == 1:
        print(ch)
        break
```

33 Longest word in sentence

```
s = "Python coding interviews are easy"
words = s.split()
print(max(words, key=len))
```

34 Count frequency of words

```
s = "python is easy python is powerful"
freq = {}

for word in s.split():
    freq[word] = freq.get(word, 0) + 1

print(freq)
```

35 Replace characters in string

```
s = "banana"
print(s.replace("a", "o"))
```

36 Remove duplicate characters

```
s = "programming"
res = ""

for ch in s:
    if ch not in res:
        res += ch

print(res)
```

37 Reverse words in sentence

```
s = "Python is easy"
print(" ".join(s.split()[::-1]))
```

38 Check rotation of string

```
s1 = "abcd"
s2 = "cdab"
```

```python
print(s2 in s1+s1)
```

39 Count uppercase and lowercase letters

```python
s = "PyThOn"
upper = lower = 0

for ch in s:
    if ch.isupper():
        upper += 1
    elif ch.islower():
        lower += 1

print("Upper:", upper, "Lower:", lower)
```

40 Find substring

```python
s = "python programming"
sub = "program"

print("Found" if sub in s else "Not Found")
```

41 Find missing number

```python
lst = [1, 2, 4, 5]
n = 5
total = n*(n+1)//2
print(total - sum(lst))
```

42 Move zeros to end

```python
lst = [1, 0, 2, 0, 3]
res = [x for x in lst if x != 0] + [0]*lst.count(0)
print(res)
```

43 Find pair with given sum

```
lst = [1, 2, 3, 4, 5]
target = 6

for i in range(len(lst)):
    for j in range(i+1, len(lst)):
        if lst[i] + lst[j] == target:
            print(lst[i], lst[j])
```

44 Rotate list by k positions

```
lst = [1, 2, 3, 4, 5]
k = 2
print(lst[k:] + lst[:k])
```

45 Intersection of two lists

```
l1 = [1, 2, 3]
l2 = [2, 3, 4]
print(list(set(l1) & set(l2)))
```

46 Union of two lists

```
l1 = [1, 2, 3]
l2 = [2, 3, 4]
print(list(set(l1) | set(l2)))
```

47 Find duplicates in list

```
lst = [1, 2, 2, 3, 3]
dups = []

for i in lst:
    if lst.count(i) > 1 and i not in dups:
        dups.append(i)
```

```python
print(dups)
```

48 Remove duplicates without using set

```python
lst = [1, 2, 2, 3]
res = []

for i in lst:
    if i not in res:
        res.append(i)

print(res)
```

49 Find Kth largest element

```python
lst = [10, 20, 30, 40]
k = 2
lst.sort(reverse=True)
print(lst[k-1])
```

50 Split list into chunks

```python
lst = [1, 2, 3, 4, 5, 6]
size = 2

for i in range(0, len(lst), size):
    print(lst[i:i+size])
```

51 Count character frequency

```python
s = "interview"
freq = {}

for ch in s:
    freq[ch] = freq.get(ch, 0) + 1
```

```python
print(freq)
```

52 Sort dictionary by values

```python
d = {'a': 3, 'b': 1, 'c': 2}
print(dict(sorted(d.items(), key=lambda x: x[1])))
```

53 Key with maximum value

```python
d = {'a': 10, 'b': 30, 'c': 20}
print(max(d, key=d.get))
```

54 Merge two dictionaries

```python
d1 = {'a': 1, 'b': 2}
d2 = {'c': 3, 'd': 4}
d1.update(d2)
print(d1)
```

55 Invert dictionary

```python
d = {'a': 1, 'b': 2}
inv = {v: k for k, v in d.items()}
print(inv)
```